

NESS Test 5: Oscillating Output Resistors

Question: In the oscillating output circuit used in "NESS Test 4" what do the 2 resistors do

Hypothesis: I believe they have something to do with timing as they are placed between the V_{cc} , $V_{Discharge}$, and Ground

Process:

1. Setup the oscillating output circuit
2. Change the 2 resistors and record changes
3. Resistor 1, ($R1$), is placed in a node with capacitor, trigger, and threshold. Going to $R2$ node
4. Resistor 2, ($R2$), is placed with discharge, $R1$. Going to V_{cc}
5. Conclude results

Data:

LED ON	LED OFF	$R1 \Omega$	$R2 \Omega$
Short	V. Short	220	220
Medium	Short	220	1000
Medium	Medium	1000	220
Long	Medium	1000	1000
V. Long	Medium	1000	10000
V. Long	Long	10000	1000
VV. Long	V. Long	10000	10000
Short	∞	1000	0
Short	VV. Short	0	1000

Conclusion:

- Originally I thought that one resistor controlled on and one off this is incorrect
- When the capacitor charges it goes through $R1 + R2$, when discharges it goes through $R2$
 - Time on = $R1 + R2$
 - Time off = $R2$
- Also $R1$ is needed or a short circuit happens